

MC211: Functions of Several Variables and PDEs

Course Placement: Functions of Several Variables and PDEs is a core course for first year students of BTech MnC program.

Course Format: It is **3 hours' lecture and 1-hour tutorial** every week.

Course Contents:

UNIT-1: (Differential Calculus) Limits, Continuity, partial derivatives, differentiability, Directional derivatives, Gradient vectors, Tangent Planes and normals, Taylor theorem (Statement only) and criteria for maxima/minima/saddle points, Lagrange Multiplier Method.

UNIT-2: (Integral Calculus) Double and Triple integrals, change of order of integration, change of variables and applications.

UNIT-3: (Vector Calculus) Basic concepts, Integration in Vector Fields: Line integrals, Surface area and Surface Integrals, Green, Gauss, Stokes Theorems and their applications.

UNIT-4: (Partial Differential Equations (PDEs)) First order partial differential equations: Linear and Quasilinear first order PDEs, method of characteristics; Second order partial differential equations: Classification of second-order PDEs, canonical form, Wave Equation, Heat Equation, Laplace Equation, D'Alemberts formula and Duhamel's principle for one dimensional wave equation, Fourier Method for IBV problem for wave and heat equation in rectangular regions, Fourier Transform and its application to solve PDEs.

Text and Reference Books:

1. George F. Simmons. Differential Equations With Applications and Historical Notes. 2nd ed. International Series in Pure & Applied Mathematics. McGraw-Hill Higher Education, 1991. ISBN: 978-0070575400.
2. George B. Thomas, Ross L. Finney, Calculus and analytical geometry, 9th edition, Pearson Publications, India, 2006.
3. Bartle and Sherbert, Introduction to real Analysis, Fourth Edition, John Wiley & Sons, 2011.
4. E. Kreyszig, Advanced Engineering Mathematics, 10th Ed., Wiley, 2015.

Assessment Method: MID-SEM + END-SEM + Project + Tutorial Assignment + Quizzes.

Course Outcomes: This course aims at building an advance understanding of calculus in multi-variable calculus and partial differential equation. After successful completion of the course the student will have the ability to

- Formulate mathematical model of real-world situations by applying abstract concepts to complex problems and solving the model either analytically or by adopting numerical methods. (**P1, P2, P3, P4, P12**).

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X								X

Lecture Schedule:

Sl. No.	Description		No. of Lectures
1	Differential Calculus		10
	1.1	Limits, Continuity, partial derivatives,	
	1.2	Differentiability, Directional derivatives,	
	1.3	Gradient vectors, Tangent Planes and Normals	
	1.4	Taylor theorem (Statement only) and criteria for maxima/minima/saddle points	
	1.5	Lagrange Multiplier Method	
2	Integral Calculus		6
	2.1	Double and Triple integrals	
	2.2	Change of order of integration	
	2.3	Change of variables and applications.	
3	Vector Calculus		12
	3.1	Basic concepts	
	3.2	Line integrals	
	3.3	Surface area and Surface Integrals	
	3.4	Green, Gauss, Stokes Theorems and their applications.	
4	Partial Differential Equations		14
	4.1	Linear and Quasilinear first order PDEs,, , Fourier Method for IBV problem for wave and heat equation in rectangular regions	
	4.2	Method of characteristics	
	4.3	Classification of Second-order PDEs and Canonical form,	
	4.4	Wave Equation, Heat Equation, Laplace Equation	
	4.5	D'Alemberts formula and Duhamel's principle for one dimensional wave equation	
	4.6	Fourier Method for IBV problem for Wave and Heat equations in rectangular regions.	
	4.7	Fourier Transform and its application to solve PDEs.	